A Transcript Grammar for the Claimed BB(7) Champion

One move, one gate, and a Coq proof of both — working notes

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Abstract

These are working notes, not a paper. We study the claimed BB(7) champion through its transcript, written one digit per step. After a fixed start of 319 digits, every digit we can observe is produced by a single move word $W(P, j)$ applied again and again, and the tape changes by one rule. A second word describes the halt, and a third a restart. Which of the two happens is decided by a gate: whether a number P is odd or even. The grammar reproduces the first 60 million digits of the run exactly, without running the machine. We prove in Coq, with no axioms, that the move holds for all values of its parameters, with its exact step count, and that the gate halts the machine when P is even. We do not prove the whole run: whether the rule applies until the end, the halting time, and the claimed bound all remain open.

1 The machine and what is claimed

`1RB0RG_1RC1RB_0RD1RZ_0LE0LD_0LF1LE_1RA1LF_1LC0RB`

The machine was found by Jeff Ketchersid in October 2026. The BusyBeaverWiki lists it as the BB(7) champion with a bound of $3 \uparrow^{20} 3$ and marks the claim as not independently verified [1]. These notes do not verify that claim, and the discoverer may already have rules equivalent to the ones below. Everything here was found from the transcript and from small test tapes.

Digits. There are 14 state–symbol pairs. Each step is written as 2·(state index)+(symbol read), with ten to thirteen written `a`, `b`, `c`, `d`:

	A	B	C	D	E	F	G
reads 0	0	2	4	6	8	a	c
reads 1	1	3	5	7	9	b	d

The digit `5` (C reads 1) is the halt. Three digits repeat in long runs: `3` (B walks right over ones), `9` and `b` (E and F walk left over ones).

2 The tape shape

At certain moments the tape is

$$1\ 0\ [\mathrm{G}\ 0]\ \ 1^{P}\ 0^{g_1}\ 1^{c_1}\ 0^{g_2}\ 1^{c_2}\ \cdots$$

with the head in state G on a 0, the two cells 1 0 to its left, and blank cells further left. We call this a *clean shape*. To the right of the head is a row of blocks of ones separated by gaps of zeros. The first block has length P. A block 1100 of the earlier notes is simply a block of two ones followed by a gap of two; we call it a *unit*.

3 The words

Name	Digits
Open	`c47698a03`
Round(n)	`2476` `9`^{n} `8a0` `3`^{n}
Build(P)	Round(4) Round(7) $\cdots$ Round($3P+1$) (P rounds)
Close(n)	`2468` `b`^{n}
Count(k)	`a1d` `3`^{k} `2468` `b`$^{k+1}$
Wait	`c476998ba1`

Every word below starts with the same piece,

$$\text{Head}(P,j) \;=\; \text{Open Build}(P)\ \text{Close}(3P{+}4)\ \text{Count}(3P{+}2)\,\text{Count}(3P{+}1)\cdots\text{Count}(3P{+}4{-}2j),$$

which has $2j-1$ countdown lines. Write $m = 3P+3-2j$.

The move word.

$$W(P,j) \;=\; \text{Head}(P,j)\ \texttt{a1d}\,\texttt{3}^m\ \text{Round}(m+1)\ \texttt{2476}\,\texttt{9}^{m+5}\,\texttt{8ba1}\ \ \text{Wait}^{\,j-1}.$$

Its length is

$$|W(P,j)| \;=\; 3P^2+27P+44+2j+(2j-1)(6P+14-2j).$$

For $j=1$ this is the five-line block of the earlier notes, of length $3P^2+33P+58$.

The halting word.

$$\begin{aligned} H(P,j) \;=\;& \text{Head}(P,j)\ \texttt{a1d}\,\texttt{3}^m\ \text{Round}(m+1)\ \text{Close}(m+5) \\ & \text{Count}(m+3)\cdots\text{Count}(0)\ \ \texttt{a1c47698b}\ (\texttt{a1c476998b})^{(m+3)/2}\ \texttt{ba1d25}. \end{aligned}$$

It ends in `5`, the only `5` of the run.

The restart word (seen for $j=1$ and $j=2$). It is $H(P,j)$ up to and including `a1c47698b`, then $(\texttt{a1c476998b})^{m/2+1}$, then a fixed ending that depends only on j (given in the script).

4 The rule

A move looks only at P, the first gap $g_1 = 2j$ and the first block $c_1 = Z$.

Move ($Z \geq 2$). The machine prints $W(P,j)$ and

$$1^P\ 0^{2j}\ 1^Z\ R \quad\longrightarrow\quad (1100)^{j-1}\ 1^{3P+8-2j}\ 00\ 1^{Z-2}\ R.$$

- *Gap of 2* ($j=1$): P becomes $3P+6$ and the block loses two. This is the tripling: $P+3$ is multiplied by 3.

- *Larger gap*: $j-1$ new units appear in front, so the first block is now a unit ($P=2$), and the old P becomes a new block of $3P+8-2j$ ones.
- *When* $Z-2=0$ the gap of two that follows joins the next gap, which therefore grows by two.

The gate ($Z=1$). There is nothing to take two from, and the parity of P decides.

P even	the machine halts and prints $H(P,j)$
P odd, and the 1 is the last cell	restart: it builds $(3P-1)/2$ fresh units and goes on

5 The run

The run is

$$\text{Start}\;\; R(13,2)\;\; W(P_1,j_1)\,W(P_2,j_2)\,W(P_3,j_3)\cdots\; H(P,j).$$

- **Start**: a fixed word of 319 digits. After it the tape is clean with $P=13$, a gap of 4 and a single 1.
- **The gate, first time**: $P=13$ is odd, so the machine restarts and builds $(3\cdot13-1)/2=19$ units. At step 3,954 the tape is clean with $P=16$, a gap of 4, nineteen units and a last single 1.
- **From then on everything is even.** If P is even then $3P+6$, $3P+8-2j$ and 2 are even, new units are blocks of two, and blocks only lose two at a time. So no block can become a single 1, and the gate is not met again until the last cell of the tape is reached. At that moment P is even and the machine halts.

So, unlike the mod-3 gate of the BB(5) champion, which is tested at every layer, this gate is tested once near the start and once at the very end.

The first steps of the real run.

Step	Move	Tape after it (blocks and gaps)
3,954	restart $R(13,2)$	$P=16$; gap 4; 19 units; 1
5,520	$W(16,2)$	$P=2$; gap 2, block 52; gap 4; 18 units; 1
5,656	$W(2,1)$	$P=12$; gap 2, block 50; gap 4; 18 units; 1
$\approx 6.06\times10^{25}$	$W(\cdot,1)$, 25 times	$P=15\cdot3^{25}-3$; gap 6; 18 units; 1

The last line is at step 60,572,642,711,595,382,146,057,206.

Why it runs so long. Each time a block is used up, two gaps join and the next move has a larger j. A gap of 2 gives tripling; a gap of 4 repeats a whole run of triplings for each two ones of the next block; and so on, one level higher for each larger gap. With nineteen units the gaps can grow to about twenty levels. This fits the shape of the claimed bound, but we have not worked out or proved any bound.

6 Tests

The script `bb7_grammar.py` writes the digits from the grammar alone.

Test	Result
First 60,000,000 digits of the real run	identical to the machine
Single moves, $P = 2 \ldots 15$ (odd and even), gaps $2 \ldots 12$	756 of 756 match
Complete small runs to the halt	169 of 169 identical

These are tests, not proofs.

7 What is proved in Coq

File `BB7Moves.v` (668 lines, Coq 8.18). `Print Assumptions` reports that `move`, `halts_gen` and `tripling_chain` are closed under the global context.

Name	Statement
`move`	The move of Section 4, for all P, $j \geq 1$, $Z \geq 2$ and any rest of the tape, whenever $m = 3P + 3 - 2j \geq 1$. P may be odd or even.
`wcost_closed`	The move takes exactly $\lvert W(P, j)\rvert$ steps.
`triple`	$j = 1$: $P \to 3P + 6$ in $3P^2 + 33P + 58$ steps.
`chain`, `tri_closed`	n triplings in a row; P ends as $(P + 3)\,3^n - 3$.
`halts_gen`	The gate: if m is odd (that is, P even) and the block after the gap is a single last 1, the machine halts.
`start`	The blank tape reaches the clean shape of step 3,954.
`at_5656`	The tape at step 5,656.
`tripling_chain`	The tape after the 25 triplings, at step 5656 + `chcost 25 12`.

The proof follows the words: one lemma for each repeated digit (`3`, `9`, `b`), one for a round, a countdown line and a waiting line, and short computations for the fixed pieces. The term `chcost 25 12` equals the 26-digit step number above by the same arithmetic done outside Coq.

Not proved.

1. **The whole run.** The move needs $m \geq 1$, that is, a gap that is not too large for P. We believe the run always satisfies this, because a small P only appears right after a move, when the gaps in front of it are new and small; but this is an argument in words. Without it there is no theorem that the machine halts and no exact halting time.

2. **The restart** for general odd P. The one in the real run is covered only by running the machine inside Coq up to step 3,954.

3. **The digit strings.** The step counts equal the lengths of the words, but the words themselves are not stated as theorems.

4. **The claimed bound.**

8 Open questions

1. Prove that the run stays inside the rule until the last cell is reached. This would give a proof of halting and an exact expression for the halting time, as was done for the BB(6) champion [3].

2. Work out the growth exactly: what is the state when each unit is used up, and what bound does that give?

3. Prove the restart word for all odd P, so that the start is 319 digits and nothing else.

4. Explain the 319-digit start. There is a clean shape already at step 26.

5. What does the machine do outside the rule ($m < 1$)?

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References

[1] BusyBeaverWiki. *BB(7).* `https://wiki.bbchallenge.org/wiki/BB(7)`, read 11 November 2026.

[2] V. Vaidyanathan. *A Proved Transcript Grammar for the BB(5) Champion.* November 2026.

[3] V. Vaidyanathan. *Exact Step Counts and an Exact Halting Time for the BB(6) Champion, Proved in Coq.* November 2026.